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Faculty of Engineering

Mechatronics Engineering Department

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MCT444s Mechatronics in Rehabilitation Technology Project Description

Closed-Loop Musculoskeletal Modelling and Simulation for Lower-Limb Robotics Rehabilitation:
Integrating CAD, OpenSim-Creator, OpenSim, and Control Strategies

Objective: This project influences the synergy of OpenSim Creator, OpenSim, and programming environments such as MATLAB or Python to develop a closed-loop controlled simulation of a musculoskeletal human model integrated with assistive lower-limb devices. These devices which may include unactuated supports, robotics/exoskeletons, or prosthetics (e.g. for below or above-knee amputees) will be designed or adapted using CAD software (e.g., SolidWorks or Inventor) and then incorporated into the musculoskeletal model through OpenSim Creator. By performing a comprehensive literature review, modeling, and simulation of assistive technologies, participants will investigate gait performance, metabolic cost, energy consumption, and trunk muscle activities. This systematic approach facilitates a deeper understanding of how such devices can improve patient/user outcomes, reduce energy expenditure, and support effective rehabilitation strategies. Students will gain practical experience in biomechanical modeling, biomechanronic device/wearable robot design, data analysis, and rehabilitation evaluation, ultimately helping to optimize exoskeleton and prosthetic design for individualized patient care.

OpenSim's built-in tools such as the Muscle Energy Meter which is a tool aims to quantify mechanical work, heat components, and overall energy consumption for both individual muscles and the entire body during walking motions. These insights in addition to using MATLAB and/or python will guide iterative design improvements and control strategies for assistive devices to minimize metabolic cost and enhance gait efficiency. By uniting cutting-edge simulation environments with evidence-based design practices, this project will contribute significantly to rehabilitation science, offering a pathway toward refined, highly effective assistive technologies and a marked improvement in the functional recovery and quality of life for individuals requiring lower-limb support.

Note: Proper citation and transparency of any models or methods sourced online (e.g., from GitHub or the OpenSim website) are mandatory. All modifications or customizations must be thoroughly documented to ensure academic integrity. If you use an online model, you must explicitly mention its source and detail any changes made—such as modifications to the model or control algorithms—and calculate energy consumption and metabolic costs accordingly. Failure to cite online resources, or to demonstrate meaningful improvements and personal contributions, may result in failing the project. All references must be clearly documented, and you must clearly identify your own work in adapting or enhancing any borrowed material.

Project Steps:

1. Literature Review

- **Biomechanics & OpenSim Tools:** Investigate existing research on biomechanics, focusing on how OpenSim, the OpenSim API, and OpenSim Creator are employed in the design and simulation of assistive technologies.
- **Assistive Devices & Control Algorithms:** Explore the state-of-the-art in lower-limb assistive devices, including unactuated supports, exoskeletons, and prosthetics (for below- or above-knee amputees). Document the control algorithms (e.g., PID, impedance control, machine learning approaches) commonly used.
- **Musculoskeletal Modeling:** Examine different musculoskeletal models and their implementations in OpenSim, noting key parameters like muscle groups, joint constraints, and trunk muscle involvement.
- **Energy & Metabolic Analysis:** Review methods for quantifying mechanical work, heat components, and total energy consumption. Familiarize yourself with OpenSim's Muscle Energy Meter and metabolics calculators to assess muscle-driven simulation for reducing metabolic cost during gait.

2. Device & Model Development

- **CAD Design:** Develop or adapt a lower-limb assistive device (soft or rigid exoskeleton/prosthetic) using CAD software (e.g., SolidWorks or Inventor).
- **Integration via OpenSim Creator:** Transfer the CAD design into OpenSim Creator, ensuring proper alignment, joint definitions, and compatibility with the musculoskeletal model.
- **Musculoskeletal Model Construction:** Build or refine a human lower-limb musculoskeletal model in OpenSim, incorporating relevant bones, joints, muscles, and tendons. If referencing any online models, cite sources and explicitly detail modifications made.
- **Control Algorithm Implementation:** Develop or adapt closed-loop control strategies (in MATLAB or Python) to regulate the device's interaction with the musculoskeletal model during simulation.

3. Simulation & Data Processing

- **Simulation Setup:** Configure the simulation in OpenSim, integrating the assistive device and the musculoskeletal model. Calibrate control parameters and run initial tests to verify stability.
- **Motion Capture (Optional):** If applicable, use motion capture data to validate or fine-tune the biomechanical model's kinematics and dynamics.
- **Metabolic & Gait Analysis:** Employ OpenSim's metabolic calculators (e.g., Muscle Energy Meter) to measure energy consumption on a muscle-by-muscle basis and for the entire model. Assess gait performance indicators such as step length, stride frequency, and joint range of motion.

4. Interpretation of Results

- **Comparison of Control Strategies:** Evaluate different control algorithms' effectiveness in improving gait efficiency, reducing metabolic cost, or enhancing stability.
- **Device Performance:** Examine how the integrated assistive device affects kinematics, kinetics, trunk muscle activities, and overall energy expenditure.
- **Model Verification:** Compare simulation outcomes against known biomechanical benchmarks or experimental data (if available) to validate model accuracy.

5. Conclusion

- **Summary of Findings:** Synthesize key results regarding the efficacy of the proposed exoskeleton/prosthetic design, highlighting its impact on gait performance, trunk stability, and metabolic cost.
- **Future Recommendations:** Recommend design or control modifications to optimize device function further, including considerations for user comfort, adaptability, and clinical feasibility.

6. References & Resources

- **Citations:** List all sources used, including academic papers, online repositories (GitHub, OpenSim website), or previously developed models.
- **Documentation of Modifications:** Clearly detail any changes made to external models, including adjustments in geometry, joint definitions, or control parameters, as well as your own original contributions.
- **Academic Integrity:** Ensure all references are properly cited. Uncited or unmodified online models may result in disqualification, so document your work to maintain transparency.

7. Report Writing, Presentation, and Video Submission

- **Detailed Report:** Prepare a comprehensive report documenting all aspects of the project, including the methodology, results, and conclusions. Ensure proper academic formatting and clear presentation of data and findings.
- **Video Presentation:** Submit a **10–15 minute voice narrated video** presentation showcasing your work and demonstrating project activities. **The video quality (both content and production) will significantly influence your grade.**
- **Project Quality:** Maintain a high standard of work. Insufficient quality or minimal effort may result in **failing** the project.
- **Presentation Session:** Present the findings to peers and/or instructors, participating actively in discussions and responding to questions.
- **Responsibility Sheet:** Include a clear breakdown of each team member's contributions.

- **CAD and Source Files:** Submit all relevant CAD models and source code files to fully document your work.

Useful Resources

1. OpenSim and OpenSim Creator

- Visit the official [OpenSim website](#) , [OpenSim repository](#) , and Opensim Creator website for documentation, examples, tutorials, and research project ideas.
- Leverage provided APIs and sample models to understand best practices in musculoskeletal simulation and biomechanical modeling.

2. Internet Search and Academic Publications

- Conduct extensive internet searches and surveys to gather information on biomechanics, musculoskeletal modeling, and robotics.
- Consult research papers, articles, and online resources in databases such as PubMed, IEEE Xplore, and Google Scholar.
- Focus on recent publications related to biomechanics, musculoskeletal modeling, and assistive devices to inform your project's design and implementation.

3. GitHub Repositories

- Explore GitHub for code examples, libraries, and projects connected to OpenSim, sensor integration, and assistive device development.
- If you incorporate any code or models from these repositories, cite them properly and document modifications you make to ensure academic integrity.

Teamwork and Evaluation Criteria

- **Team Formation:** Teams of up to five or six students are permitted. Each member must clearly document their individual responsibilities and the percentage of their contribution.
- **Use of External Models:** Adopting models directly from the internet without proper referencing or without adding significant modifications/improvements will lead to project failure.
- **Assessment Criteria:** Grading will take into account the significance of the proposed idea, scientific rigor, methodology, attained results, teamwork, task distribution, quality of the written report, presentation effectiveness, and the video submission that showcases project activities.
- **Referencing Requirements:** All sources, including online models and code, must be appropriately cited and acknowledged. Any external models used must undergo substantial customization to avoid academic misconduct.

- **Practical Learning Experience:** This project serves as an opportunity for students to apply biomechatronics, simulation, and assistive device control knowledge in a hands-on manner, thereby enhancing critical thinking and problem-solving abilities.
- **Proposal Submission:** Each team must submit a 2-page proposal via email to m.hamdi@eng.asu.edu.eg by **March 16, 2026**, outlining the project idea (exoskeleton or prosthetic for the lower limb), objectives, methodology, detailed plans, team member roles, and used tools.
- **Learning Resources:** For additional tutorials on OpenSim, consult YouTube, OpenSim's official documentation and website, as well as the HCM Lab YouTube channel <https://www.youtube.com/watch?v=kFeXeC9Two0&list=PL49pWqgsffJb3uwGRYzf3fqsKDg1KzhWB>.